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MIL-STD-1898(AT)
Notice 1
12 August 1991

MILITARY STANDARD

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5. DETAILED REQUIREMENTS

5.1 Metallurgical standards.

5.1.1 Applicability. This section provides guidance in establishing metallurgical requirements relative to material inclusions and inspection criteria for surface (case) hardening of ferrous gears.

5.1.1.1 Classification. Gear classification shall conform to those designated on engineering drawings. Established levels of inspection shall be in accordance with the following categories of gear areas:

a. Class 1 gears. These are gears for heavy duty service which could approach and/or operate under loads reaching the physical limits or capability of the material. They must have the ability to withstand high bending and compressive stresses, shock or impact loads and resist wear.

b. Class 2 gears. These gears are for medium duty service which could operate for a major portion of the life under loads substantially below the physical properties of the material and must have the ability to withstand moderate bending and compressive stresses, moderate shock or impact loads and resist wear.

c. Class 3 gears. The severity of service that these gears are subjected to is considered to be light as compared to Class 1 and 2 gears. They are relatively lightly loaded and used in application where resistance to wear is the primary factor in obtaining adequate gear life.

5.1.1.2 Gear areas. Figures 1a and 1b depict specific areas of gears and are defined as area A, B and C relative to the compressive stresses, bending, shock or impact loads anticipated.

a. Class 1. Areas shall be in accordance with figure 1a and 1b.

b. Class 2. Gear teeth and splines shall be areas A or B. The remainder of the gear shall be considered area C.

c. Class 3. All areas shall be considered gear area C.

5.1.1.3 Acceptance criteria. The criteria to be used for determining the acceptability of gears which have imperfections are specified in tables I, II and III, and as specified below.

a. Acceptable indications: Magnetic particle indications associated with nonmetallic inclusions which do not exceed the following:

1. Nonrateable indications: Any one indication, 0.06 inch or less in size which does not occur in axial strings.

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2. Rateable indications: Indications which conform to the requirements of the applicable tables for the following gears:

   a. Class 1 - Table I.

   b. Class 2 - Table II.

   c. Class 3 - Table III.

b. Unacceptable indications are specified as follows:

   1. Cracks, laps, bursts, etc., and grinding checks.

   2. Through wall indications.

   3. Indications breaking over a machined edge on Class I and II gears.

5.1.1.4 Hardness. Gear hardness shall be as specified on the engineering drawing.

5.1.2 Test methods.

5.1.2.1 Test method. Visual and magnetic particle inspection shall be performed to determine conformance to the acceptance criteria specified in 5.1.1.3 for the class of gear specified. Magnetic particle inspection shall be conducted in accordance with MIL-STD-1949.

5.1.2.2 Hardness testing. Upon the completion of all heat treat and case hardening operations, gear class specifications shall conform to the following criteria and shall be measured as shown on figure 2.

a. Class 1:

   1. Tooth surface hardness measurements shall be taken on a minimum of 3 gear teeth located approximately 120° apart on the gear tooth area.

   2. Tooth surface hardness measurements shall be taken with the Rockwell 15 N scale or equally sensitive measuring device, and suitable fixtures, on the tooth profile of finished machined gears.

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3. Case hardness measurements on carburized or nitrided gears or case hardness measurements on the tooth outside diameter shall be taken using the Rockwell 30N scale, or equally sensitive measuring device, and suitable fixture.

b. Class 2 and Class 3:

1. Tooth surface hardness measurements shall be performed on the load side of each tooth face at the pitch diameter using the Rockwell 15N scale or equally sensitive measuring device, unless otherwise specified on engineering drawings.

2. Spiral and bevel gears shall conform to case hardness requirements specified on the engineering drawings.

c. Core hardness measurements on all gear classes shall be taken on a sectioned gear tooth or a sectioned test specimen tooth.

5.1.2.3 Rework criteria. Gear areas containing indications may be reworked by blending, provided the minimum dimensions and surface finish requirements of the engineering drawings are met. Except for the specific methods of rework stated herein, all other methods used for general rework shall require approval of the Material Review Board, prior to use.

5.1.2.4 Blending. Indications shall be smoothly blended into the surrounding area by means of a carbide burring bit or equivalent tool. All blending shall conform to the limits specified in table I, and figures 1a and 1b.

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TABLE I. Acceptance limits for class 1 gears.

Gear Area A Any indication (rateable or unrateable) shall subject the gear to rejection.
Gear Area B 1. Surface indications 0.25 inch or less and sub-surface indications 0.37 inch or less shall be acceptable. 2. If the distance of closest approach between indications in an axial direction is less than the length of the larger indication (rateable or unrateable), the indications (2 or more) shall be evaluated as a single indication. 3. Parallel surface or sub-surface indications shall be separated by 0.25 inch minimum. 4. Nor more than 2 acceptable indications shall be permitted in a 1.50 inch diameter circle. 5. Not more than 4 indications shall be permitted per gear in the B areas.
Gear Area C 1. Indications 0.50 inch or less shall be acceptable. 2. If the distance of closest approach between indications in an axial direction is less than the length of the larger indication (rateable or unrateable), the indications (2 or more) shall be evaluated as a single indication. 3. Parallel indications shall be separated by 0.25 inch minimum. 4. Not more than 4 indications shall be permitted per part in the C areas.

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1. All tooth profile lines must be within the area defined by the band shown on the involute diagram.
2. All tooth profile lines must be smooth and uniform. Concave, wavy or irregular tooth profile lines are not acceptable when the cavity, wave, or irregularity represents a rate of departure from the measured mean profile line in excess of the rate of departure specified on the gear drawing (figure 3 which illustrates both acceptable and unacceptable profile charts).

d. In matching the actual measured involute profile to the tolerance band, the start of the actual measured involute profile diagram may be placed at any point on the True Involute Form (TIF) diameter as represented on the chart within the confines of the band at that diameter.

5.2.2.4 Tooth parallelism error. To comply with 5.2.1.3 the tooth parallelism and crown limits specified on the gear drawing apply only to the gear loaded face(s). On gear drawings where the acceptable crown is not given, it should be taken as one half (1/2) the acceptable parallelism error. All tooth surfaces, loaded or unloaded, not otherwise controlled on the gear drawing shall be parallel with the axis within 0.002 inches per inch. In making the parallelism check, the gear must be supported in relation to the axis. The indicator must travel in a line parallel to the axis and contact the tooth surface in the area of the pitch point (rolling pitch diameter). Figure 3 illustrates this technique. Because of machining operations performed on gears after the teeth have been finished, it is possible for a finished gear to have an axis of rotation that is not parallel to the axis of the base cylinder of the teeth (see figure 10).

a. If the parallelism of the surfaces of each tooth is checked with respect to the rotational axis of the gear, and the measurements plotted along a line representing the teeth, the results will approximate a sine wave as shown in figure 11.
b. Figure 11 illustrates the parallelism check of a 15 tooth gear. It can be readily seen that some of the gear teeth are within the parallelism tolerance. However, if any three equally spaced gear teeth are examined at least one of the teeth is in fact out of tolerance. For example look at teeth numbers 1, 6, and 11. Teeth number 1 and 6 are almost within tolerance while tooth number 11 is 0.001 inches over the tolerance. Therefore when checking parallelism, check 3 gear teeth approximately 120° apart. Only straight or crowned gear tooth surfaces are acceptable, with no concavity allowed. Figures 12a and 12b illustrate acceptable and unacceptable tooth surface.

5.2.2.5 Tooth spacing tolerance.

5.2.2.5.1 Tooth spacing tolerance - base pitch. To check for tooth spacing errors of 5.2.1.4, tooth to tooth error readings may be taken by using a device consisting of a pair of fingers, one fixed and the other

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actuating a dial indicator. Both are mounted on a movable block or slide with means to accurately return the slide to the same position. The position of the slide, when the readings are being taken, should be such that the fingers contact the teeth on a line tangent to the base circle. A spring tension device is desirable to keep a uniform pressure of the gear against the fixed extender (see figure 5). The indicator is set at zero for the first pair of teeth and all readings between successive pairs of teeth are relative to the first reading. Table IV illustrates the method of showing readings recorded for a twelve tooth gear.

a It is important, when interpreting the reading, to note that the readings are not errors. The difference between successive readings denotes spacing error. The reliability of this check is limited by the sensitivity of the equipment and the skill of the operator. Slight misinterpretation of the dial indicator readings are as often in the minus direction as they are in the plus direction, and the two cancel each other out when a large number of teeth are checked.

b. Table IV illustrates this method by showing readings marked down for a twelve (12) tooth gear.

NOTE: The difference between successive readings denotes spacing error. The reliability of this check is limited to the sensitivity of the equipment and skill of the operator.

5.2.2.5.2 Tooth spacing tolerance - pitch diameter. While the pitch for cylindrical gears is the theoretical length of a circular are, actual checking is accomplished by measuring a straight line, or chordal dimension at pitch diameter "A", as illustrated in figure 10a. Pitch measuring instruments employ a fixed finger and stop for consistent positioning on successive pairs of teeth, and a movable finger which displays pitch variations on a dial indicator or chart recorder. Readings are influenced by profile variations and runout of the gear as mounted in the checking instrument (see figure 10c). (Tooth-to-tooth spacing is a measurement of three (3) adjacent profile per figure 10b. Tabulate data as in table IV) Tolerance values are the same as those for base pitch spacing.

5.2.2.5.3 Accumulated tooth spacing tolerance. Accumulated tooth spacing measuring devices include index heads, dividing plates, optical polygons, theodolites, etc. Profile variation and runout may affect these readings, depending upon the device and method. These devices may also be used to determine spacing variations. Angular-position devices, such as the Follows No. 8 Microdex, and the MAAG SP-60/ES-401 Pitch Tester with Process Computer, have the capability of measuring index. Both the Bellows and the MAAG can present the data in chart form. The MAAG can also present the maximum index variation digitally and automatically, which makes calculating the index, on scaling from a chart unnecessary. The index variation can also be presented digitally, step by step, together with the maximum variation, giving the extreme positive and negative variations, as well.

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a. Interpretation of data. Figure 14 compares tabulated and graphic results of pitch (tooth-to-tooth) and index (angular position) measurements on the same hypothetical gear. This figure, for illustrative purposes only, uses a rack which is considered a gear with an infinite number of teeth. The figure shows the method of calculating pitch from measured values of spacing or angular index.

5.2.2.6 Tooth thickness error (normally measured over two pins). To check tooth spacing error of 5.2.1.5, the pin size to be used in making measurements over, or between, two pins should be listed on the gear drawing. Only standard gear measuring pins should be used. They are correct at 68°F. The measurements between, or over, two pins must be within the tolerance specified on the gear drawing over the full circumference of the gear. For gears with even numbers of teeth, the pins should be placed in diametrically opposite tooth spaces. For gears with odd numbers of teeth, the second pin should be placed in the space either side of the tooth that is diametrically opposite the first pin. The measurements that are listed on the gear drawing take into account the odd or even numbers of teeth; no adjustment in the values is required.

5.2.2.7 Runout. Runout of a gear may be measured by observing, or graphically recording, the center-distance variation when rotated about its rotational axis tightly in mesh with a master gear, excluding the effect of any center-distance variation caused by profile errors, interference, etc. figures 13a, 13b, and 13c illustrate runout caused by eccentricity, out of round, and scallops respectively (see 5.2.1.6).

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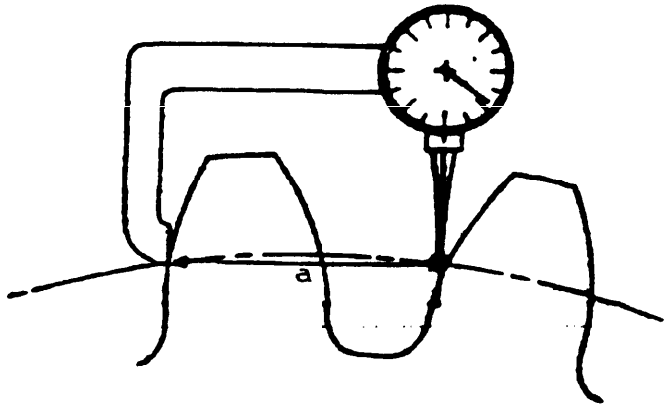


FIGURE 10a. Pitch - a measurement of two adjacent profiles.

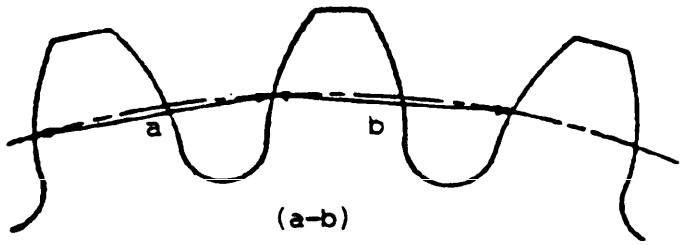


FIGURE 10. Tooth-to-tooth spacing - a measurement of three adjacent profiles.

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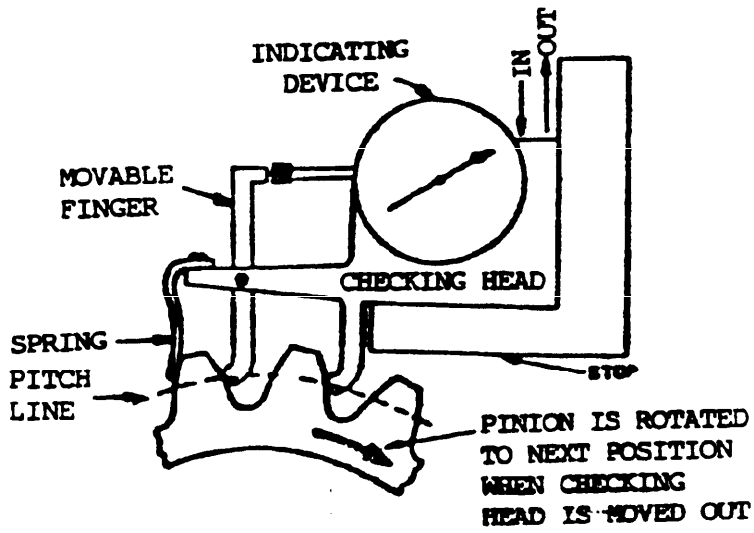


FIGURE 10c. Schematic arrangement of a pitch measuring instrument.

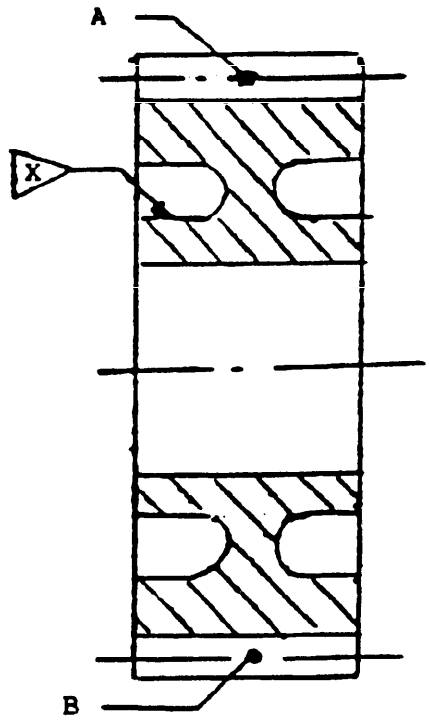
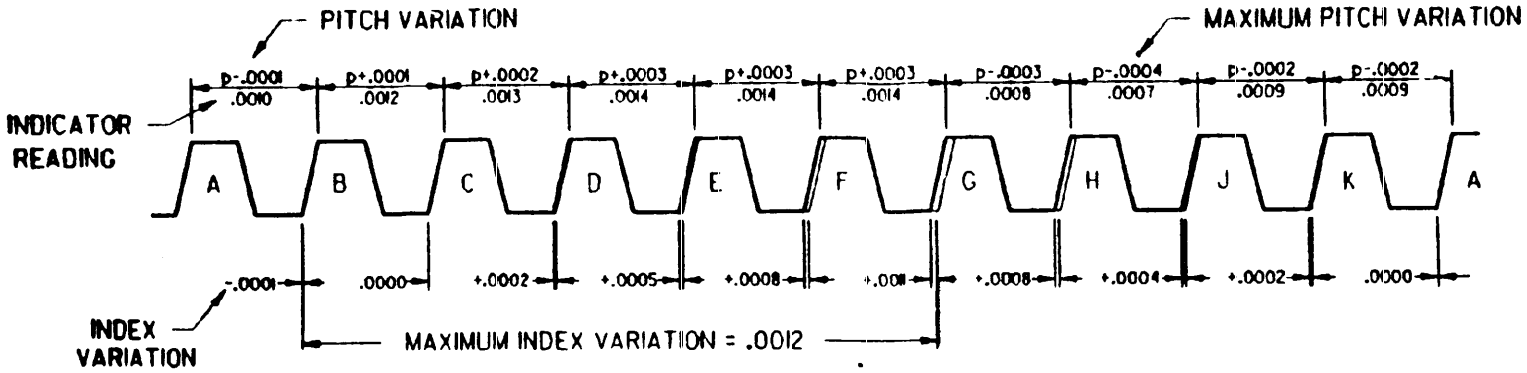


FIGURE 10d. Identification of tooth sides.

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HEAVY LINE REPRESENTS PERFECT GEAR

1. SET INDICATOR TO .0010 ON FIRST PAIR OF TEETH AND ENTER AS FIRST READING IN COLUMN NO. 1.
2. RECORD SUCCESSIVE TOOTH-TO-TOOTH INDICATOR READINGS AROUND GEAR (COLUMN NO. 1).
3. SPACING VARIATION (COLUMN NO. 2) IS THE DIFFERENCE BETWEEN SUCCESSIVE INDICATOR READINGS IN COLUMN NO. 1.
4. TOTAL ALL INDICATOR READINGS AND DIVIDE BY NUMBER OF TEETH TO FIND AVERAGE SPACING WHICH REPRESENTS THE CORRECT PITCH.
5. PITCH VARIATION (COLUMN NO. 3) IS THE DIFFERENCE BETWEEN EACH INDICATOR READING OF COLUMN NO. 1 AND THE AVERAGE SPACING.
6. TO FIND INDEX VARIATION (COLUMN NO. 4) ADD SUCCESSIVE PITCH VARIATION FROM COLUMN NO. 3.
7. TO FIND MAXIMUM INDEX VARIATION, ALGEBRAICALLY SUBTRACT THE MAXIMUM AND MINIMUM VALUES IN COLUMN NO. 4.

TEETH	INDICATOR READINGS	SPACING VARIATION	PITCH VARIATION	INDEX VARIATION	
A - B	.0010	.0002	-.0001	-.0001	-.0001
B - C	.0012	.0001	+.0001	.0000	
C - D	.0013	.0001	+.0002	+.0002	
D - E	.0014	.0000	+.0003	+.0005	
E - F	.0014	.0000	+.0003	+.0008	
F - G	.0014	.0006 MAX.	+.0003	+.0011	(-1 + .0011) .0012 = MAX. INDEX VARIATION
G - H	.0008	.0001	-.0003	+.0008	
H - J	.0007	.0002	-.0004 MAX. PITCH VARIATION	+.0004	
J - K	.0009	.0000	-.0002	+.0002	
K - A	.0009	.0001	-.0002	.0000	

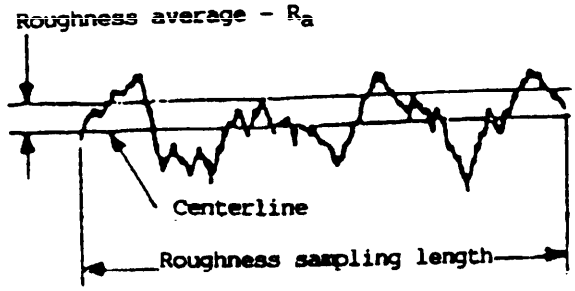
10 | .0110 CHECK: TOTAL = 0 CHECK: LAST NUMBER = 0

.0011 = AVERAGE INDICATOR READINGS (FOR CORRECT PITCH "P")

Figure 14. Comparison of pitch and index measurements on the same gear.

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5.2.2.8.1 Surface texture - roughness. The active tooth surface, over which roughness measurements are taken, is the active face of the tooth as shown in figure 21, not including fillet, root, chamfer, edge rounds and outside diameter. Measurements of roughness are made in microinches, millionths of an inch (0.000001), of the arithmetic average deviation from centerline:



$$R_a = \frac{1}{L} \int_{x=0}^{x=L} |y| dx$$

where R_a = arithmetic average deviation from the centerline.
y = ordinate of the curve of the profile and
L = the sampling length.

a. On earlier drawings, the surface texture requirement for the load side of the tooth is indicated by a check mark with a number, for example $\overset{8}{\checkmark}$. On later drawings, the surface texture requirement for the finished gear is in the form of a tooth for the involute trace, and a diagonally trailing line for the lead trace. As in the figure 23 example, roughness is the plane of rotation (involute) shall be between 0.000005 minimum and 0.000016 max; roughness along the face (lead) shall be held to 0.000016 max. The simple check mark symbol shall continue to be used for other gear tooth requirements such as the root, loaded face before honing, and the unloaded face.

b. The tip spherical radius of the tracer head shall be 400 ± 120 microinches new, or 400 ± 200 microinches used. The stylus force shall be 1.5 grams.

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5.2.2.8.2 Surface texture - waviness. On earlier drawings, unacceptable waviness is defined with the following note: "Rate of departure from the measured mean profile must not exceed X.XXX inches in 25 percent of the total roll from the true involute diameter to the outside diameter." The actual measured tooth profile may be completely within the tolerance band, but may exceed the maximum acceptable rate of departure (slope). This condition is shown in figure 25. Later drawings require waviness to be measured on an involute checker at a magnification equating the space between the horizontal chart lines to 0.00005 inch, as shown in figure 24. In the example, the longest wave shows the waviness to be "B = 0.000080". In the figure 23 symbol for surface texture, the waviness requirements are indicated with a "W".

5.2.2.8.3 Acceptance standard (Ra 8 finish). To achieve a finish of Ra 8 microinches or better, it is necessary to hone after grinding.

a. Surface roughness measurements shall be conducted using a Perthometer, model C5D or equivalent. The cut off shall be 0.010 inch, the traverse length 0.060 inch, and the sensitivity 50.

b. Surface roughness measurements shall be taken on five teeth, with spaces between teeth approximately equal. Three of the five readings shall be taken at mid-plane, the fourth reading near the aft end, and the fifth reading near the forward end when checking in the involute direction. The surface finish requirements is for any direction on the tooth. Inspection shall determine if roughness is a maximum in lead or involute direction, and live teeth shall be inspected accordingly.

c. The average roughness of the five readings shall not exceed the specified roughness and the reading on any single tooth shall not exceed nine microinches.

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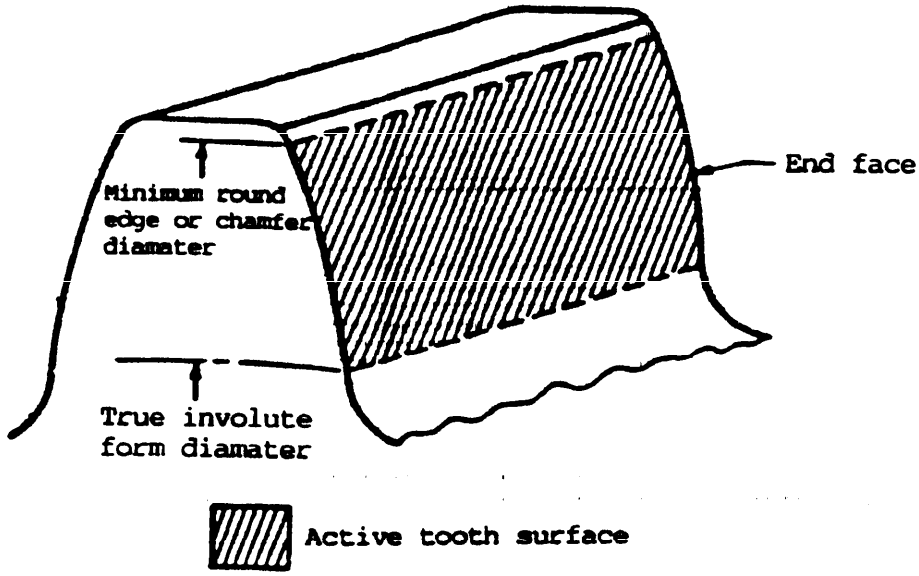


FIGURE 21. Active tooth surface.

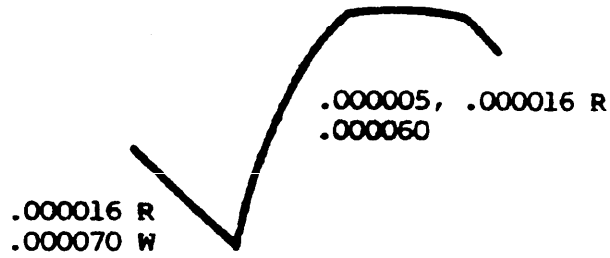


FIGURE 23. Example of surface texture specification.

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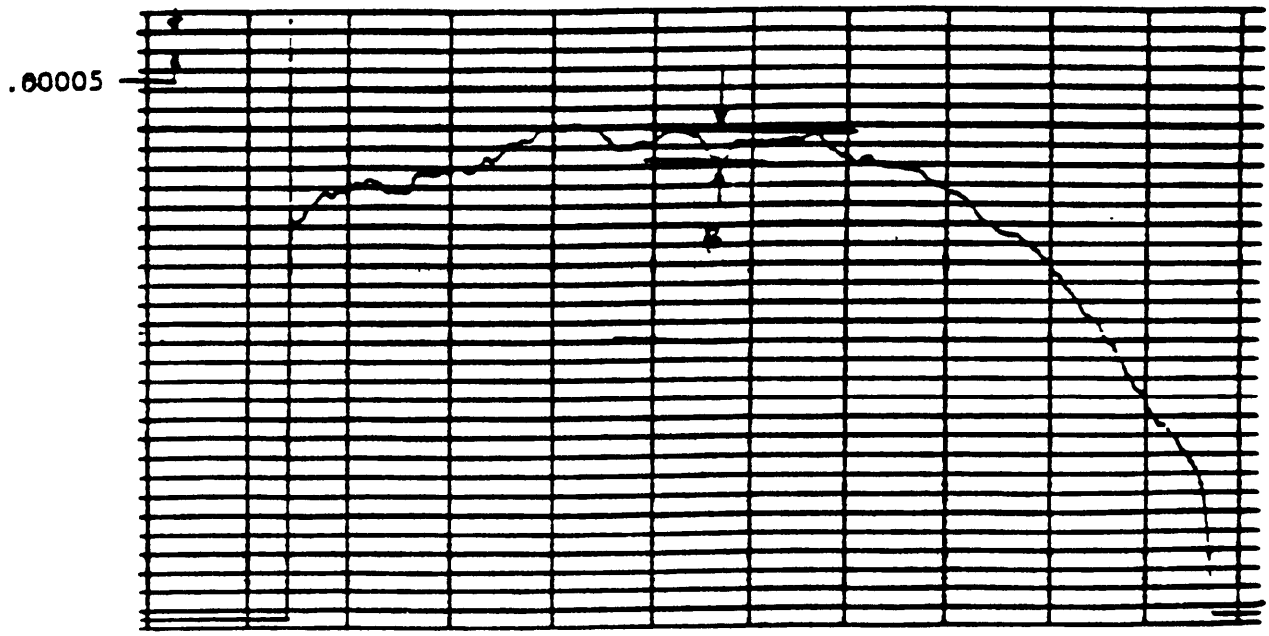


FIGURE 24. Waviness measurement.

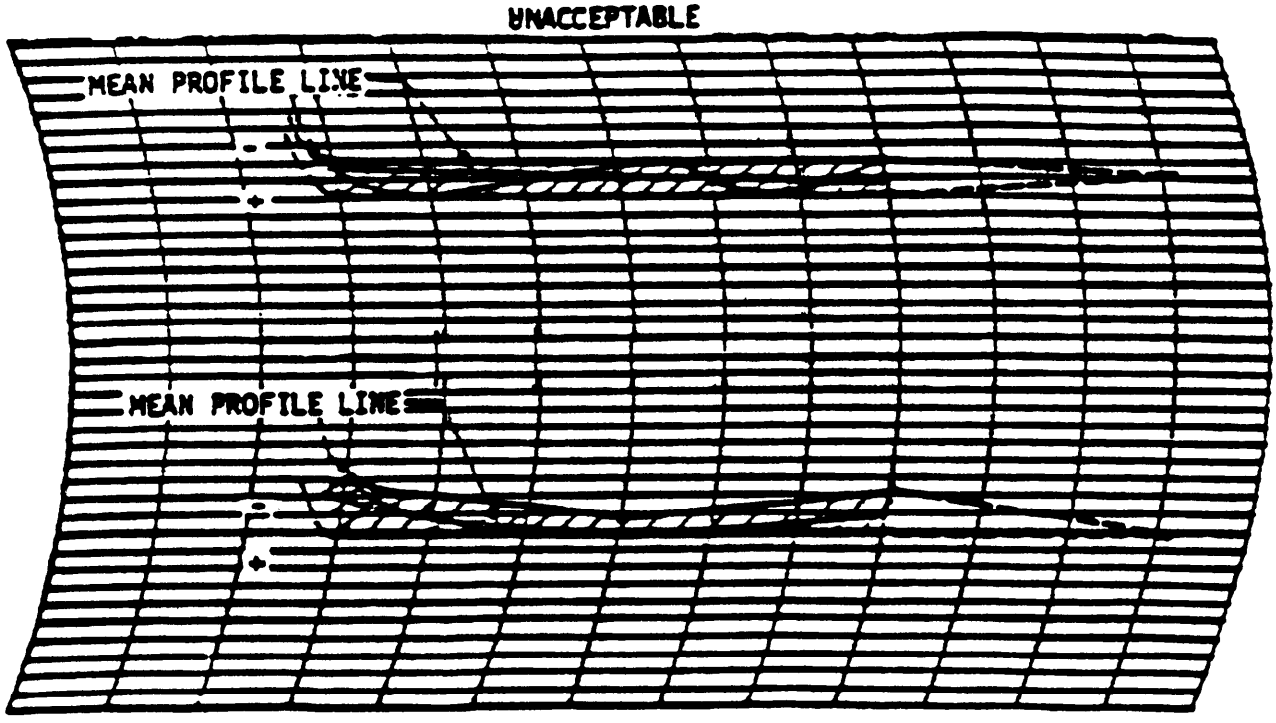


FIGURE 25. Unacceptable condition for surface texture - waviness.

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5.3.6.3.2 Noise and smoothness of operation. The test for noise and smoothness is a check of gear operation under normal running conditions in the final mountings. This test should be combined with the tooth bearing check described above. While the gears are running in mesh for the tooth bearing test, the operator should listen for any unusual noise types and levels. Any other indications of rough action should also be recorded.

5.3.6.3.3 Location of top land and back face. These surfaces may be checked as shown in either figure 24 (part A or B). However, inspection may develop other methods (for example, comparator methods) so long as they are adequate to insure conformance of the gear to drawing requirements. The drawing for each particular gear furnishes sufficient basic dimensions to properly position and zero the dial indicators (see figure 24, part A) or to properly position the contact points (see figure 24, part B). Figure 25 shows the specific basic dimensions given on the drawing. It should be noted that the basic diameter given is a dimension which is concentric with the X diameter. It is important that the set-up of the checking device be made to satisfy this requirement. Once having accomplished this set up, the method of figure 24, part A, checks the surface locations by direct reading of the dial indicators, while the method of figure 24, part B, makes use of a feeler gage to be inserted between the contact points and the production gear. For either method it is not necessary that surfaces be checked along their entire length. Two points reasonably close to either end (lengthwise) of the particular surface will be sufficient.

NOTE: Diameter X will be specified on the drawing. It will be selected as that diameter upon which the gear is mounted in final assembly.

5.3.6.3.4 Backlash. The measurement of backlash is made in the gear testing machine. The gear being inspected is assembled in mesh with its mating gear. With the gear fixed, the rotation of the production gear is measured using a dial indicator as shown in figure 26. This measurement is taken normal to the active tooth surface at the extreme heel of the gear. This is a suggested method, other methods may be employed so long as they are equivalent to this. Care should be taken to convert all readings to normal backlash should the method employed not indicate this value.

5.3.6.3.5 Runout. Runout is measured in a Gleason concentricity tester or its equivalent. To make this measurement, the production gear is held in a position which allows it to rotate about its axis. With one tooth held against a fixed stop, a dial indicator reading is made on a tooth diametrically opposite. The contact points are mounted on a swinging bracket which can be lifted while the gear is turned from tooth to tooth. All teeth are checked in this manner, and the measured runout is equal to one-half the difference between the high and low readings obtained.

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5.3.6.3.5.1 Tooth spacing. The spacing of individual teeth of a bevel gear is measured in a Gleason tooth spacing tester or equivalent. This device has two contact points which can be adjusted to contact corresponding sides of adjacent teeth. One contact point is used as a fixed stop, and the other actuates a dial indicator reading to ten-thousandths inch. The two contact points are carried on a swinging bracket which can be lifted out of position and then returned to permit the checking of successive teeth in the gear. The indicator is set to zero for the first pair of teeth. All subsequent readings are therefore relative to the first pair of teeth selected. In interpreting the readings, it is important to note that the readings are not the tooth-to-tooth spacing errors. These errors are the differences between successive readings. The reliability of this check is limited by the sensitivity of the equipment and the skill of the operator. However, experience shows that the slight misinterpretations of the dial indicator readings are as often in the minus direction as they are in the plus direction, thereby cancelling each other when a reasonable number of teeth are checked. To illustrate the method, the readings for a ten tooth gear are given in table VI.

5.3.6.3.5.2 Hardness check. Hardness checks shall be made to determine conformance to drawing requirements. A minimum of three (3) samples shall be selected from each heat treat lot for determination of effective case depth, core hardness and surface hardness. Samples shall be processed by the same heat treat process procedure used for production parts. Inspection and hardness tests shall be performed in the areas shown in 5.1.

5.3.6.3.5.2.1 Failure. Failure of any one (1) test sample shall be cause for rejection of the entire representative production lot. Items produced subsequent to failure of the test sample shall not be submitted for acceptance until corrective action has eliminated the cause of the failure. Acceptance of heat treat samples subsequent to failure shall not resume until three (3) consecutive samples have successfully met specified test requirements.

5.3.6.3.5.3 Finish. Verify that the surface finishes specified on applicable part drawings conform to specified requirements, and that there are no burrs or sharp edges on any part of the gears.

5.3.6.3.5.4 Marking. Verify that the gears are permanently marked as specified on applicable part drawings and that the bevel gear sets are serialized and maintained as matched sets.

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TABLE VI. Readings in ten-thousandths of an inch.

Column 1 Tooth pair	Column 2 Indicator reading	Column 3 Difference between successive readings of column 2
1	0	+2
2	+2	+1
3	+3	+1
4	+4	-2
5	+2	-2
6	0	+1
7	+1	-3
8	-2	+2
9	0	+3
10	+3	-3
1	0	SUM 0

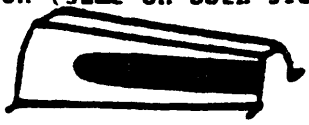
Since the gear has a full number of teeth, any spacings which are too large must be offset by others which are too small. Therefore, the sum of the all the differences (column 3) must be zero. If uncompensated errors are made in reading the indicator, the sum of column 3 will not equal zero. If the sum of column 3 deviates from zero by more than 0.0002 inch, the check must be repeated. The largest number (regardless of sign) in column 3 is the maximum tooth spacing error.

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(a) Pinion (same on both sides)



(b) Gear (same on both sides)

Correct Tooth Bearing



(a) Pinion (same on both sides)



(b) Gear (same on both sides)

Out of Position Bearing
Cause: Pinion to close to cone apex.
Remedy: Move pinion away from cone apex.

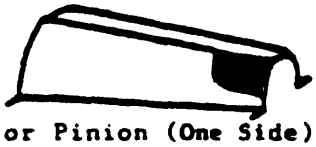


(a) Pinion (same on both sides)



(b) Gear (same on both sides)

Out of Position Bearing
Cause: Pinion too far from cone apex.
Remedy: Move pinion toward cone apex.



(a) Gear or Pinion (One Side)



(b) Gear or Pinion (Opposite Side)

Cross Bearing
Cause: Shafts do not intersect.
Remedy: Line up shafts



(a) Gear or Pinion (Both Sides)



(b) Gear or Pinion (Both Sides)

Toe or Heel Bearing
Cause:
Toe Bearing: Shaft angle too large.
Heel Bearing: Shaft angle too small.
Remedy: Correct the shaft angle.

Note:

These sketches show the possible tooth bearings for a pair of correctly-made straight bevel gears which are not of the coniflex type. The correct tooth bearing shown above applies to all straight bevel gears of this type regardless of differing service conditions.

FIGURE 35. Tooth bearings on straight bevel gears and pinions.

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6. NOTES

6.1 Supersession data. This document supersedes MIL-STD-1898(AT), dated 13 July 1983; AVCO Lycoming Division specifications E2152A, dated 26 April 1982; P6805D, dated 8 June 1988; P6920B, dated 27 September 1979; and P7800E, dated 14 July 1986.

6.2 Subject term (key word) listing.

Bevel gears
Gears
Spur gears

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 3020-A006)

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